UNIFIED MECHANICS · VOLUME D · TRAVERSAL PAPER 00

# The Retained Fraction and the Discarded Half

*Two places where this volume touches the Corpus. One supplies a number the Corpus was owed. The other returns something the Corpus threw away in its fifth movement and has been missing ever since.*

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On the grades, which are the Corpus's and are used here unchanged. PROVED follows from stated definitions or assumptions. CONDITIONAL follows once a named condition is granted. PROPOSED is a physical reading offered for calculation and falsification. OPEN names what can be formulated but not yet settled. Both computations below reproduce from the Corpus reproducibility layer, scripts/shields_field_check.py and scripts/05_06_cosmology.py.

SECTION ONE

## What this paper is for

The papers in this volume were written outward. They take one claim, that a completed traversal is not undone by identifying its endpoint with its origin, and carry it into Goedel spacetime, into the AdS/CFT domain, into cosmology, into the string worldsheet, and finally into a field defined on pairs of points. None of them was written to serve the Corpus, and that is why they are worth reading: they had to survive contact with named external results rather than with the framework's own vocabulary.

But two of them come back. This paper states where, and states it before the volume begins rather than leaving a reader to notice it in a footnote five papers in. One hook supplies a quantity the Corpus lists as owed. The other returns an object the Corpus discarded deliberately in Movement Five and has been working without ever since.

SECTION TWO

## The first hook: the invariant is the retained fraction

The Rate Across the Scale finds one condition reproducing both ends of cosmic history. The quantity that matters is not a value but a rate, and the invariant governing the rate is the second moment of the acceleration parameter over the first.

**eta_H = <epsilon squared> / <epsilon>**

Two properties of that invariant do the work. The cosmological constant cancels out of it exactly, contributing nothing to either moment, so the invariant is independent of how much of it there is. And holding it constant reproduces the standard cosmological model as an exact solution rather than a limiting case. Both are established in that paper without any framework input.

At the late end the invariant is three halves, which is the matter fixed point, standard, and owed to nobody. At the early end it is measured from the spectral tilt and the tensor bound. And that is where this volume meets the Corpus, because the Corpus already carries a retained fraction per crossing.

```
corpus retention per crossing   1 - r^3    =  0.970491502813


n_s    predicted  1 + ln(1 - r^3)  =  0.970047
n_s    measured                   =  0.9649  +- 0.0042      pull 1.23 sigma


eta_H predicted  ln(1 - r^3)/2    = -0.014976
eta_H measured                    = -0.0176 +- 0.0021      pull 1.25 sigma
```

The inflationary invariant is the logarithm of the Corpus's own retention over one crossing, halved. Nothing was fitted to reach that; r is fixed by the retained fraction of a single act of self-description and has been since Movement Four, and the logarithm is what a per-crossing retention becomes when read as a rate per e-fold.

### What this does to the Corpus's fifth obligation

The Formal Register lists O5 as the integral joining the growth of the record to the measured expansion history, with the zeroth-order clause proved and the residual open. That description is now wrong in the way an obligation is wrong when it has been attacked successfully but not finished.

**Supplied** The expansion history in full, as the exact constant-invariant trajectory, at both ends, with the cosmological constant cancelling identically out of the invariant that governs it.

**Supplied** The early-time value of the invariant, from the Corpus's own retained fraction, at a pull of about one and a quarter standard deviations.

**Not owed** The late-time value. It is the matter fixed point and is standard.

**Still open** Why the invariant is constant. That is the whole of the residue, and it is one question rather than a missing functional.

> *O5 is not discharged and this paper does not claim it. It is reduced to a single question of the same shape that O2 was reduced to before it was closed: not a missing object, but one stated condition wanting a reason. The numerical agreement is PROPOSED, as every identification of a framework quantity with a measured one is in this corpus.*

SECTION THREE

## The live stake this volume carries

Holding the invariant constant is not free. The flow equation then determines the acceleration parameter, and that feeds back into the tilt at second order, giving the running of the spectral index in observables alone.

**alpha = -8 epsilon (epsilon - eta_H) ~ - r (1 - n_s) / 4**

Negative in sign, and bounded in magnitude below about three and a half parts in ten thousand, falling linearly with the tensor-to-scalar ratio. The measurement today is consistent in sign and comfortably consistent in magnitude, and it is roughly twenty times less sensitive than the prediction requires. CMB-S4 does not reach it either, falling short by about a factor of six. A full twenty-one centimetre survey does.

This is the volume's own condition of defeat and it should be read as one. A measured running larger than a few parts in ten thousand, or positive in sign, ends the constant-invariant condition at the inflationary end and with it everything Section Two just claimed. The Corpus asked for a number that could be checked and found wrong. This is that number, it is not yet measurable, and it will be.

SECTION FOUR

## The second hook: what Movement Five discarded

Least-Cost Holonomy and Co-Distinction Fields defines an object with no standard name: an antisymmetric two-point function on a domain, whose two orientations are read as lighter and darker, generated together by one relation rather than existing as two substances. It is a one-cochain, and whether it reduces to a value at a point is a cohomology question with a clean answer: it does exactly when the cocycle condition holds on every triple.

The obvious guess is that this restates the Corpus's two-sided reading, which says a self-description reads its mixed relation in both directions. The guess is wrong, and the way it fails is the useful part.

```
A3 doubles the self-pairing, which is how T9 gets even norms   ->  symmetric
antisymmetry forces D(x,x) = -D(x,x) = 0                        ->  diagonal vanishes
```

They are opposite on the diagonal. No identification is available and none should be claimed. But a two-point structure on the carrier splits into a symmetric part and an antisymmetric part, and only one of them has ever been used.

```
two-point structure on a rank-8 carrier :  64 components
   symmetric part, governed by A3       :  36
   antisymmetric part                   :  28
```

Movement Five discards the second one, and it says so in the sentence that does it. The two mixed terms of the squared channel expansion are the same pair read in opposite order, and the Corpus counts them once on the stated ground that reading direction is irrelevant. That decision produced the three weights and everything downstream of them, and it is not being questioned here. But it was a decision, it threw something away, and the co-distinction field is exactly what was thrown.

SECTION FIVE

## Where the discarded half lands

The Corpus now forces a decay on a rank-three subsystem with a rank-five survivor. Decompose the antisymmetric part under that split and it does not scatter.

```
Lambda^2 of the decayed 3-block    :  3
Lambda^2 of the surviving 5-block  : 10
cross terms, three by five         : 15
                          total   28
```

The surviving ten is the degree-two sector of the exterior algebra on the five complex directions, and that sector is not an abstract ten. It is a specific list of wedges, and the Corpus has already built it.

```
Q    (colour ^ weak)   6    c1^w1  c1^w2  c2^w1  c2^w2  c3^w1  c3^w2
u^c  (colour ^ colour) 3    c1^c2  c1^c3  c2^c3
```

```
e^c  (weak ^ weak)     1    w1^w2
                      10
```

Those are the same wedges, state for state, that carry the quark doublet, the up-type singlet and the positron in the Corpus's own construction of one generation. The co-distinction field restricted to what survives the decay is half of a Standard Model generation. Not analogous to it. The same object.

> *PROVED as an arithmetic and combinatorial statement: the antisymmetric part of the carrier's two-point structure decomposes as three, ten and fifteen under the decay, and the ten is the degree-two exterior sector the Corpus already constructed. PROPOSED as the claim that the co-distinction field of this volume IS that antisymmetric part rather than merely sharing its shape. Certified by scripts/shields_field_check.py.*

SECTION SIX

## Why that is the right place for it to land

There is a reason this is not a coincidence of dimensions, and it is old enough to be unarguable. A symmetric two-point form gives a metric. An antisymmetric one gives a connection, and therefore holonomy, and therefore curvature.

The Corpus takes the symmetric route throughout. The two-sided reading gives an even lattice, the even lattice gives the carrier, and the inherited quadratic form gives the Lorentzian signature. Every geometric result in the Corpus descends from the symmetric side, and none of them touches the antisymmetric side because Movement Five had already set it aside.

And the Corpus's gravitational statement has always been about the other side without having the object to say it with. Gravity is the surviving part of carrier nonclosure. The Source Sector proves that a co-distinction field gravitates only when it is not potentiated, which is to say only when its cocycle fails. Nonclosure and cocycle failure are one condition written in two vocabularies, and the second vocabulary has a theorem attached and an obstruction class to name.

**the symmetric half gave the metric; the antisymmetric half is the connection**

SECTION SEVEN

## What is not being claimed

**Not claimed** That O5 is discharged. It is reduced to one question, why the invariant is constant, and that question is not answered anywhere in this volume.

**Not claimed** That the numerical agreement at one and a quarter standard deviations is a derivation of the spectral tilt. It is a framework quantity landing near a measured one with nothing fitted, which is a test passed and is not the same thing.

**Not claimed** That the co-distinction field is the antisymmetric part rather than a structure of the same shape. Establishing that requires showing the field's values on carrier pairs are the components the split produces, and this paper shows the dimensions and the wedges match, which is necessary and is not sufficient.

**Not claimed** That any of this supplies a coupling, a mass, or the Yukawa hierarchy. The Corpus owes those and this volume does not pay them.

What is claimed is narrower and, if it holds, more useful than any of the above. The traversal programme did not import anything into the framework. One of its papers supplied a number the framework was owed, from a fraction the framework already carried. Another recovered a structure the framework had explicitly set aside, and the structure came back landing on results derived afterwards and independently. Neither outcome was arranged, because neither could have been: Movement Five was written before the decay was known, and the retained fraction was fixed before anyone looked at a spectral tilt.